

Marios Saleptsis

ELECTRICAL ENGINEER PH.D

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Milano, Italy

EXPERIENCE

September 2025 – February 2026

Visiting Researcher – Polytechnique de Montreal, Canada

I conducted collaborative research in power energy systems, with a focus on optimization and machine learning applications for decision-making in power systems. Specializing in end-to-end learning in gradient-based and gradient-free methods for optimizing the forecasting process in energy systems.

February 2025 - Current

Teaching Assistant – Politecnico di Milano, Italy

Lecturer on the master's course "Electric Conversion from Green Sources of Energy" and "Fundamental Theory of Electric and Magnetic Circuits" with the duties of course material preparation, oral presentation, weekly lectures, numerical exercises solving in topics of power electronics, static converters, and electrical machines. Tutor on seminars on AI/ML – Power Systems and master's thesis supervisor.

November 2023 – Current

Doctoral Researcher – Politecnico di Milano, Italy

I served as a researcher on several national and European funded projects closely aligned with my doctoral research. I contributed to the Centro Nazionale per la Mobilità Sostenibile (MOST), the Management Energy Systems for Smart Island (MESSI), and the Network for Energy Sustainable Technologies (NEST) project. In addition, I held the role of lead engineer in the research and development of the forecasting division of DomOpti, a spin-off startup of the Politecnico di Milano. Moreover, I am a lead member of the MG²Lab research team of Politecnico di Milano.

February 2023 – January 2024

EU Project Manager – Cluster of Bioeconomy and Environment, Greece

I coordinated Horizon Europe research projects on renewable energy and advanced energy systems, managing cross-national partner collaboration, project timelines, and EU funding compliance. Contributed to proposal development, work package structuring, and periodic reporting to ensure alignment with technical milestones and budgetary requirements.

November 2020 – September 2021

Electrical Project Engineer – EcoEnergy SA, Greece

Contributed to the design, construction, and commissioning of photovoltaic plants, combining office-based engineering work with on-site supervision. Coordinated supply chain logistics across multiple projects, resolved technical incidents, and ensured delivery on schedule and within budget.

EDUCATION

November 2023 – Current (Expected November 2026)

Doctor of Philosophy (Ph.D) in Electrical Engineering – Politecnico di Milano, Italy

Developing ML-driven forecasting and optimization models for optimal microgrid scheduling, integrating day-ahead PV, load, and electricity price forecasting with MILP/LP scheduling under market, operational, and physical system constraints. Validating all the methods on the MG²Lab microgrid testbed.

September 2016 – June 2023

Diploma in Electrical and Computer Engineering – University of Western Macedonia, Greece

Integrated master's degree 300 ECTS, dedicated in electrical/energy engineering and computer science. Specializing in the power systems sector of the department.

September 2021 – February 2022

Semester Abroad – Polytechnic University of Madrid, Spain

Completed a semester abroad through the EU ERASMUS exchange programme, conducting hands-on work in the IoT Laboratory and Renewable Energies Laboratory.

LANGUAGES

- **Greek:** Native
- **English:** Full working Proficiency (C2)
- **Spanish:** Intermediate (B1)
- **Italian:** Intermediate (B1)

SKILLS

- **Expertise:** AI modelling, Electricity Market (DAM), Power Systems, Forecasting, Optimization
- **Programming:** Python (NumPy, Pandas, Scikit-learn, Matplotlib), SQL, C
- **Machine Learning:** PyTorch, TensorFlow/Keras, Machine and deep learning models expertise, time-series forecasting, AI and DL in power systems
- **Mathematical Optimization:** CVXPY, Pyomo, Gurobi, MILP/LP formulations, cvxpylayers
- **Tools:** GitHub, Docker, GitLab, TS database, Grafana, Linux, Homer, DigSilent,

PUBLICATIONS

Published: <https://scholar.google.com/citations?user=uJFeDVoAAAAJ&hl=en>

Under Review/Accepted:

- M. Salepsis, R. Ramaschi, A. Fusco, M. Mussetta, S. Leva, "Bridging the Gap between Predict-Then-Optimize and Decision-Focused Learning: A Cost-Aware PV Forecasting Framework," IEEE Transactions on Power Systems
- M. Salepsis, A. Lessage-Landry, M. Mussetta, S. Leva, "Game-theoretic Decision-focused Learning for Microgrid Scheduling under Multi-source Uncertainty," IEEE Transactions on Smart Grid
- M. Salepsis, M. Mussetta, S. Leva, "A Novel Hierarchical Stacked Framework for Short-term Load Forecasting: An Extreme Learning Machine and Extreme Gradient Boosting Ensemble Method," IEEE Transactions on Industry Applications
- M. Salepsis, A. Lessage-Landry, M. Mussetta, S. Leva, "Decision-Focused Learning for Optimal BESS Scheduling Under Load and PV Uncertainty in Microgrids," Energy, Elsevier
- M. Salepsis, M. Mussetta, S. Leva, "A Stacked Ensemble Approach for Probabilistic PV Forecasting: Combining Physical Hybrid Neural Networks with Natural Gradient Boosting," ELECTRIMACS 2026, Palermo, Italy, May 2026.
- R. Ramaschi, M. Salepsis, E. Ogliari, S. Leva, "Towards a Peer-Based Probabilistic Framework for Photovoltaic Performance Monitoring with Aggregated Data," ELECTRIMACS 2026, Palermo, Italy, May 2026.